

Two decades of research on business intelligence system adoption, utilization and success – A systematic literature review

NoorUl Ain^{a,*}, Giovanni Vaia^a, William H. DeLone^b, Mehwish Waheed^c

^a Department of Management, Università Ca' Foscari Venezia, S. Giobbe, 873, 30121, Italy

^b Kogod School of Business at American University, 4400 Massachusetts Ave NW, Washington, DC, USA

^c Center for Higher Education, Technische Universität, Emil-Figge-Straße 50, 44227 Dortmund, Germany

ARTICLE INFO

Keywords:

Business intelligence system
Adoption
Utilization
Success
Systematic literature review

ABSTRACT

In the recent era of technological advances and hyper-competition, business intelligence (BI) systems have attracted significant attention from executives and decision makers due to their ability to provide complex and competitive information inputs for the decision process. Following the world of practice, research into the adoption, utilization and success of BI systems has grown substantially over the past two decades. The literature suggests that organizations have largely failed to capture the benefits of BI systems to their full extent and are seeking ways to leverage value from the implemented systems. However, prior studies do not have any comprehensive study that discusses the issues and challenges related to adoption, utilization and success of BI systems. In this study, using a systematic literature review, we present comprehensive knowledge about what has been found in the domain of BI system adoption, utilization and success. A total of 111 peer-reviewed studies, covering three categories – adoption, utilization and success – published between 2000 and 2019, were selected. The findings present the research methods, underpinning theories and key factors employed to study BI system adoption, utilization and success. In addition, the review identified the key issues related to BI adoption, utilization and success and highlighted the areas that have attracted more or less attention. This study also suggests future directions for researchers and practitioners in terms of unexplored themes that may help organizations to obtain value from BI systems.

1. Introduction

Advances in organizational information systems and technologies led to the emergence of business intelligence systems in the late 1990s [1,2]. A business intelligence (BI) system is commonly known as a suite of technological solutions [3] that facilitates organizations to amass, integrate and analyse vast stocks of data in order to understand their opportunities, strengths and weaknesses [4]. BI is an information system that supports decision processes by i) facilitating more aggregation, systematic integration and management of unstructured and structured data, ii) dealing with a huge amount of data (e.g. “Big Data”), iii) providing end-users with increased processing capabilities to discover new knowledge [5], and iv) offering analysis solutions, ad hoc queries, reporting and forecasting [6,7]. According to Clark, Jones and Armstrong [8] BI systems are quite close to the original DSS concept, since they extend the categories of users and support a wider variety of decisions. Certainly, as part of the broad call of management support systems (MSS), they are designed to reduce uncertainty in the

decision-making process, and support the decision maker efficiently and effectively [8].

A BI system is a combination of tools, such as a data warehouse, online analytical processing (OLAP) and dashboards. A data warehouse gathers accurate, clean and detailed data from multiple sources for in depth analysis [9], whereas online analytical processing (OLAP) supports multidimensional analysis in real time and enables users to apply operations such as aggregation, filtering, roll up and drill down for details (e.g. products, customers, times, country, region) and pivoting [8,10]. Furthermore, dashboard servers as the front-end application for data visualization and performance management. It enables users to create graphs, charts, widgets and ad hoc reports and decision makers to track the key performance indicators of the business [8].

With increased competition from both online and traditional businesses, these technological solutions have become extremely important for organizations to improve their managerial practices and performance as well as their products and services [11,12]. The BI market increased worldwide by about 7.3% in 2017, with revenues up to \$18.3

* Corresponding author.

E-mail addresses: noor.ulain@unive.it (N. Ain), g.vaia@unive.it (G. Vaia), wdelone@american.edu (W.H. DeLone), mahwish.phd@gmail.com (M. Waheed).

billion, and it is expected to reach \$22.8 billion by the end of 2020 [13]. However, despite the growing investments and great market expansion, the evidence has suggested that many organizations fail to reap benefits from the implemented BI systems [14]. Above the 70% of BI projects fail to yield the expected returns [15] or result in few or no benefits for organizations [16]. Organizations are in a continuous struggle to find the best way to leverage value from BI systems and to make their implementation a success [17].

Scholars and practitioners are still debating strategic and tactical approaches to the successful adoption and use of BI systems, producing hundreds of publications through different media. However, a limited number of studies have attempted to synthesize this existing body of knowledge. For instance, Jourdan, Rainer and Marshall [18] reviewed BI studies from 1997 to 2006 with a focus on research strategies and methods. Similarly, Fitriana, Eriyatno and Djatna [19] reviewed the progress made in BI studies from 2000 to 2011. They discussed the most popular research approaches – the single approach and the integrated approach – used within BI studies. The analysis revealed that 50% of the research focused on a single approach and discussed the definition, theory, methodology and architecture of BI systems, whereas the rest of the research focused on BI integration with other areas, like supply chain management, customer relationship management and artificial intelligence. Trieu [12] analysed BI studies from 2000 to 2015 to understand the processes through which organizations can attain value from BI systems.

Meanwhile, an extensive stream of BI research has been conducted in the past two decades. In fact, hundreds of single studies have been published in the categories of adoption, utilization or success at the organizational and individual levels [20–23]. However, to date, no synthesis exists on the three categories of the adoption, utilization and success of BI systems as a means of ascertaining the current status of the BI research. Therefore, through a systematic literature review (SLR), this study aims to: (i) *comprehensively report on areas of investigation (adoption, use and success), theories/framework/models, key factors and challenges and (ii) identify knowledge gaps that need further investigation and suggest opportunities for future research.*

The remainder of the paper is structured as follows. The second section outlines the research questions and research methodology to explain the process of finding relevant articles from leading databases. The third section presents the findings with respect to the proposed research questions. The fourth section discusses the limitations of the study. The fifth section outlines the theoretical and practical implications of our findings and the conclusions.

2. Methodology

This study undertook SLR based on the guidelines suggested by Kitchenham [24]. Initially, a comprehensive review protocol was developed to guide the SLR. The aim of a review protocol is to minimize the likelihood of bias in a study, and it is thus considered as an essential element of SLR [24]. The protocol provided a detailed plan for the systematic review by specifying the approaches to be followed and the quality measures or conditions to apply while selecting the literature [25]. It involved the following stages: the identification of research, research questions, search procedure, criteria for study selection, quality assessment, data extraction process and data synthesis [26]. The

previous section of this study described the first stage, that is, the identification of the research, while the following sections describe the remaining steps.

2.1. Research questions

The research on BI system AUS is still progressing, however, no effort has been made to systematically review existing research in BI research domain from AUS perspective. Given that, this study aims to firstly investigate the main areas of investigation within BI AUS research domain. This provides a rich overview of BI AUS studies from 2000 to 2019 indicating the current state of BI AUS research. Secondly, the study aims to report on areas of investigation (adoption, use and success), theories/framework/models, key factors adopted, challenges and knowledge gaps. To achieve the objectives, the following five research questions are introduced.

Since the domain of this research is BI AUS, it is helpful to understand how much research attention has been given to each of these categories; therefore, the following research question was developed:

RQ1: *What is the main area of investigation in BI system AUS studies; adoption, use or success?*

Researchers need to better understand the various information systems and organizational theories used in BI studies in order compare and contrast BI research findings and to create a cumulative knowledge of BI AUS; therefore, the following research question was developed:

RQ2: *What are theories/frameworks/models adopted by studies regarding BI system AUS?*

Researchers and practitioners are interested in the organizational, informational and user factors that drive adoption, use and success; therefore, the following research question was developed:

RQ3: *What are the key factors identified in BI system AUS studies?*

Researchers and practitioners are also interested in the challenges faced when implementing BI: therefore, the following research question was developed:

RQ4: *What challenges are faced by organizations with respect to BI system AUS?*

A better understanding of knowledge gaps in BI research will open new windows for researchers and practitioners to understand the areas where further research is required; therefore, the following research question was developed:

RQ5: *What are the knowledge gaps within current BI system AUS research?*

The following section presents the criteria for BI studies inclusion and exclusion in the SLR.

2.2. Study inclusion/exclusion criteria

Inclusion/exclusion criteria were applied to ensure that the studies were relevant and within the boundaries of the research objective [26]. The inclusion criteria were applied to full-length peer-reviewed studies and conference papers related to BI systems AUS research, as depicted in the following Table 1. Furthermore, studies that were not available in full, book chapters, discussion notes, editorials and reports, highly technical articles and duplicated studies were excluded from the review list. Table 1 represents the detailed inclusion/exclusion criteria.

Table 1
Study inclusion/exclusion criteria.

Inclusion criteria	Exclusion criteria
• Studies published between the period of 2000–2019 • Studied within the domain of BI systems AUS research • Full-length peer reviewed studies • Published in the English language • Available in selected electronic databases	• Studied outside the domain of BI system AUS research • Studied with highly technical perspective, books, discussions, reports and non-scholarly work • No full-length peer reviewed studies • Not published in the English language • Duplicated

2.3. Search and selection procedure

To explore the relevant material for this review, the search was conducted through both an automated and a manual search [24]. First, the automated search was primarily based on search terms or keywords and was performed as an electronic search using electronic databases [27]. Accordingly, ten leading databases were located through Google Scholar, encompassing journals in the field of information systems, management information systems, operations management, business, management, social science interdisciplinary and information science, as shown in Table 2. The selected databases were deemed to be highly relevant, providing a comprehensive census of the literature in the BI field.

The search terms of interest, as shown in Table 2, were applied to the identified electronic databases to assess the relevant literature. The selected search terms included business intelligence system, data warehouse, online analytical processing (OLAP) and dashboards, and the adoption, use and success terms were used interchangeably. Apart from simple search strings, the Boolean operators AND/OR were applied to collect as many results as possible. The year 2000 was chosen as the starting year, as the first few academic articles related to BI were found to have been published in that year [28,29].

Additionally, a manual search was performed by adopting the backward and forward approach [27] to ensure the completeness of the systematic search. In the former approach, citations of the identified articles were reviewed to trace additional references, and, in the latter approach, the collected references were used further to identify relevant articles. Along with the empirical studies that had a key focus on BI system AUS, this review also takes into account conceptual research that contributed to the BI system AUS literature.

On completion of the search process, the study identified 612 articles that were potentially relevant to the BI domain. The inclusion/exclusion criteria discussed in the previous section and demonstrated in Table 1 were applied to the results to find the most relevant studies. Initially, the title and abstract of each article were scanned. Despite having search terms appearing in the titles or abstracts, some studies were not conducted in the BI system context and were thus found to be irrelevant to this review. Therefore, in the first step, 413 articles were excluded. This reduced the number of studies to 199, which were further filtered by skimming the full contents of the articles to ensure their relevance to BI system AUS. This resulted in the elimination of 76 irrelevant articles from the review list, leaving 123 articles relevant to this study's subject. The following Fig. 1 presents the article selection and retention process.

2.4. Quality assessment

In the final step, the quality of all 123 identified studies was assessed using Quality Assessment Criteria. The aim of the quality assessment (QA) was to make decisions regarding the overall quality of the identified studies to ensure the value of their findings and interpretations [24,30]. In doing so, the five (QA criteria) questions listed below were developed to evaluate the remaining studies:

Q1: Does the research topic addressed pertain to BI system adoption, utilization and success?

Q2: Is the context of the research clear?

Q3: Does the research adequately delineate the methodology?

Q4: Is the data collection procedure adequately explained?

Q5: Is the approach used for data analysis appropriately explained in the research?

To assess the quality level, three quality rankings – “high”, “medium” and “low” – were used for each QA criterion [30]. A study was assigned a score of 1 for a quality criterion if it completely satisfied that criterion. Similarly, a study was assigned a rating of 0.5 if it partially satisfied a quality criterion, and a score of 0 was given when a study did not satisfy a quality criterion. In this study, the highest

Table 2
Search procedure.

Years	2000–2019
Search terms	“Business intelligence”; “business intelligence system”; “antecedents of business intelligence system success”; “business intelligence system success, BI system acceptance, intelligence system adoption, intelligence system use”; “data warehouse ‘adoption’ and/or ‘use’, OLAP ‘adoption’ and/or ‘use’, dashboard ‘adoption’ and/or ‘use’”; “business intelligence system” and/or “success” and/or “adoption” and/or “acceptance” and/or “use”
Sample journals	- MIS Quarterly - MISQ Executive - Decision Support Systems - Journal of Management Information Systems - European Journal of Information Systems - Journal of Information Systems - Information & Management - Communications of ACM - Information Systems Management - Journal of Management Information Systems - Expert Systems with Applications - Behaviour & Information Technology - Computers in Human Behavior - Journal of Strategic Information Systems - Behaviour & Information Technology - Information Development - Telematics and Informatics - Highly relevant articles from other journals
Databases	Emerald insight, ScienceDirect, EBSCOhost, ProQuest, Wiley Online library, IEEE Xplore, JSTOR Archive, Taylor & Francis Online, Sage Journals, Springer-Link, Web of Science

possible rating was considered to be 5 (i.e. 5×1) with regard to the 5 QA criteria, while the lowest possible rating was 0 (i.e. 5×0). Based on the coding scheme, a study was considered to be of high quality: if > 3 e.g. 3.5, medium quality: if < 3 and > 1 e.g. 1.5, 2, and 2.5, low quality: if < 1 e.g. 0.5. as exemplified in Table 3.¹

In total, 97 studies were found to be of high quality (79%), whereas 14 studies (11%) were assigned to the group of medium-quality studies. The 12 studies representing low quality (quality score < 1) were excluded from the review list, leaving 111 studies for the SLR.

3. Results

3.1. Data extraction and synthesis process

After the selection of 111 relevant studies, the next step was to extract data from them. In this step, a data extraction form was developed to record information from the identified studies to ensure the completeness of the data collection [24]. Prior studies have suggested several elements to extract data from literature [31–34]. Accordingly, this review considered elements such as the study identifier, for example, a study ID (SID), study reference and year, study type, study objective, study context, tool/system used, framework/theory, key factors, research method and country/region, as shown in Table 4.

To record the data for each element accurately, the content of each article was carefully reviewed to extract data from each study using Microsoft Excel spreadsheets and Endnote.

Table 5,² shown above, exemplifies the recording of data in the data extraction form. Along with the above-mentioned data extraction elements, articles were also scanned for challenges faced by organizations

¹ Complete Quality Assessment Criteria (QAC) is available at the following link: <https://drive.google.com/file/d/1QHa3ZJt25yCIDQ4XG5dSULAxITsDUw-4/view?usp=sharing>.

² Complete data extraction form is available at the following link: https://drive.google.com/file/d/1XL9d_uMgXUUhq-D9JgA5vc_DwLh3kTxX-/view?usp=sharing.

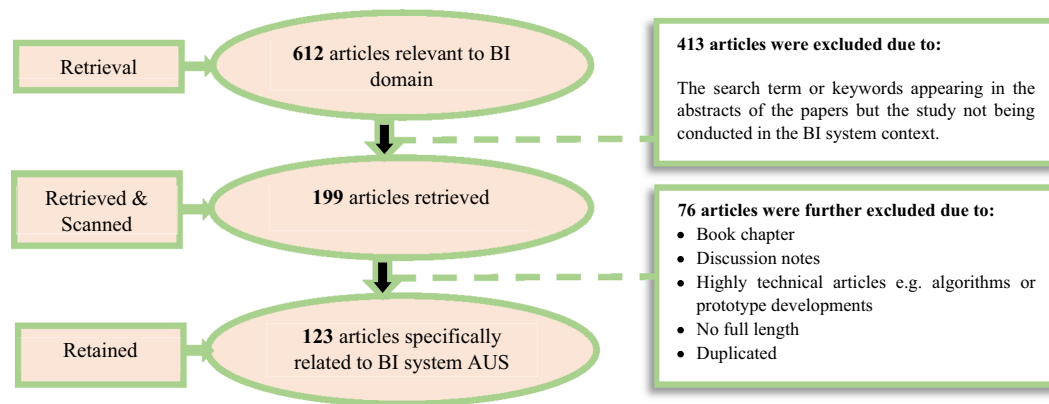


Fig. 1. Articles selection and retention process

Table 3
Quality Assessment Criteria (QAC).

Study ID	Q1	Q2	Q3	Q4	Q5	Total
B1	1	1	0.5	1	0.5	4
B2	1	1	1	1	1	5

regarding BI system AUS to investigate the fourth research questions. Table 13 presents the extracted challenges in detail.

3.2. Descriptive findings

For this study, a total of 111 studies were selected to be reviewed. Among the 111 studies 27 were published in the highest publication outlets according to the Association of Business School (ABS) ranking. Table 6 shows the number of BI system AUS articles published in journals ranked as 4*, 4 & 3 by the ABS. Note that among these top ranked journals Decision Support Systems has published the most articles i.e. six.

Following the data extraction process, the information obtained from the identified studies was analysed further. The following sections present a descriptive analysis of the results.

3.2.1. Chronological distribution of the studies

The research on BI system AUS have gained prominence over the last two decades. Fig. 2 represents the distribution of all the studies from the period 2000 to 2019. It was noticed that there were only a few publications in the years 2000, 2001 and 2002, while the number increased to two in 2003, three in 2004, three in 2005 and two in 2006 and 2007. Likewise, a total of 11 studies were found in the period 2008–2010.

From 2011 to 2019, there was a significant increase in the number of studies: a total of 87 articles were published. In addition, the data analysis revealed a change in the focus from BI systems' components to

BI systems over the period 2000–2019. In the earlier years, research primarily focused on either data warehouses (DW) or online analytical processing (OLAP), or in some cases both tools were studied together, as shown in Table 7.

In the later years, the focus of research shifted to BI systems. From 2009 to 2019, a total of 86 studies referred to BI systems.

3.2.2. Research methods

Several research approaches, including the qualitative and quantitative methods, have been adopted in BI system AUS research. The distribution of research types, namely conceptual, qualitative, quantitative and mixed methods (quantitative + qualitative), is presented in Fig. 3.

Fig. 3 reveals that a quantitative research approach has been used by the majority of BI system AUS research. Out of 111 studies, 56% adopted a quantitative approach, as shown in Fig. 3. The survey method was chosen frequently in these studies (62 studies), as shown in Table 8. Mail or web-based questionnaires were used to collect data, and the majority of respondents were managers, BI professionals and executives who were using BI systems [36,46,47]. On the other hand, 19% used a qualitative approach, as depicted in Fig. 3. However, Table 8 presents a total count of 27 studies employing qualitative research methods - the difference in number is due to, in some cases, two or more research methods (e.g. case study and interviews) being used in a single study. In addition, it was found that eight studies used interviews, one used the Delphi method and two used observations, whereas 12 studies adopted the case study method (Table 8). In-depth interviews and the Delphi method were utilized to express employees' perceptions of BI systems use [6,48].

Of the total, 11% of studies used mixed methods, as shown in Fig. 3.

3.2.3. Studies coverage by geographical regions

The 111 studies selected for this SLR covered at least 39 countries, as shown in Table 9. The number of articles from the United States was

Table 4
Elements of the data extraction form.

Data extraction elements	Description
Study ID	As a study identifier
Study reference and year	To present the authors and publication year
Study type	To identify studies as journal articles or conference papers
Study objective	To describe the aim of the study
Study context/topic	To state the study theme, for example adoption/utilization/success
Tool/system used	To identify the investigated tool, for example, a data warehouse or OLAP, and system, for example, a BI system
Framework/model used	To identify the framework/model adopted by BI AUS studies
Key factors	To identify the key factors adopted to study AUS
Research method	To identify the research approach, for example quantitative, qualitative or mixed
Country/region	To identify the countries or regions in which the study was carried out

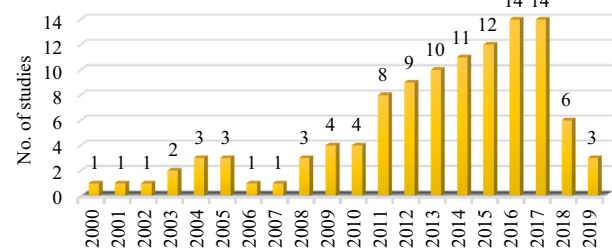
Table 5
Data extraction form.

SID	Author (s)	Study type	Objective (s)	Topics	Tool/system	Key theories/frameworks/ models	Key factors	Sample	DC method	Country
B1	[28]	Journal (I&M)	Identified factors of end-user satisfaction with data warehouses	Adoption	Data ware house	NA	Support provided to end-users, accuracy, format and preciseness, fulfillment of end-user needs	42 business managers	Survey	United States

Note: *DC = data collection, *NA = not available.

Table 6
Publications by journals.

Journals	Association of Business Schools (ABS) ranking		
	4*	4	3
Information Systems Research	1		
MIS Quarterly	1		
The Accounting Review	1		
Journal of Management Information Systems		3	
Journal of the Association for Information Systems		1	
Decision Support Systems			6
Expert Systems with Applications			4
Journal of Strategic Information Systems			2
European Journal of Information Systems			1
Computers in Human Behaviour			1
Information Systems Journal			1
Totals	3	4	20

**Fig. 2.** Publications by years from 2000 to 2019.

relatively high with 24 studies. Taiwan contributed the second greatest number of studies with 17, followed by Australia, China and South Africa (identified in twelve, eight and seven studies, respectively).

In addition, among the developed economies, Germany contributed five studies, France three studies, the United Kingdom two studies, Canada two studies, and Italy one study. On the other hand, among the emerging economies, two studies were found from Iran, Thailand, Peru and India, respectively, as shown in Table 9. In summary, the majority of research focusing on BI system AUS was conducted in the United States, Taiwan, Australia and China. At the same time BI systems are a truly global business solution.

3.2.4. Distribution of studies by sector

The analysis revealed that multiple sectors were frequently addressed by BI system AUS research (40% of the research), such as government services, transportation, insurance, communications, health care, banking, agriculture, construction and professional services [49,50]. We found fewer studies in the finance (6%), telecommunications (4%), education (2%), manufacturing (3%), health (2%), retail (3%) and semiconductor sectors (3%), as shown in Fig. 4.

However, 23% of the research did not mention or clearly define the sectors in which the study was conducted.

3.3. Descriptive findings with respect to proposed research questions

3.3.1. Areas of investigation

The 111 studies were reviewed to determine their primary area of investigation, that is, the adoption, utilization or success of BI systems. Hence, the research topic categorization is aligned with these three areas of investigation based on the primary focus of each study. We defined the adoption, utilization and success concepts as follows (Table 10):

From the categorization we can state that the topic BI system

Table 7
BI system/BI tools adoption since 2000–2019.

Years	Year system/tool studies					
	DW	OLAP	Combined (DW and OLAP)	Dashboards	Combined (DW and BI system)	BI system
2000–2004	6	1	2	0	0	1
2005–2008	3	2	0	0	0	3
2009–2012	1	0	0	0	0	22
2013–2015	2	0	0	0	0	31
2015–2017	1	0	0	1	0	26
2018–2019	0	0	0	0	2	7
Total	13	3	2	1	2	90

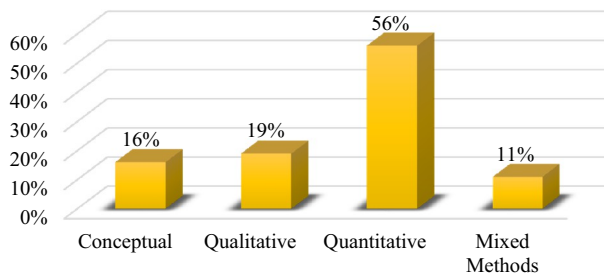


Fig. 3. Distribution of research approaches (2000–2019).

Table 8
Research methods.

Research approach	Research method	Article Count
Qualitative approach	Case study	12
	Interviews	9
	Delphi	1
	Observations	2
	Secondary data	3
Quantitative approach	Survey	62
Mixed method approach	Mixed method	11
Others	Not available	11

Table 9
Studies published by regions.

Country	Article Count	Country	Article Count	Country	Article Count
United States	24	United Kingdom	2	Japan	1
Taiwan	17	India	2	Hong Kong	1
Australia	12	Peru	2	Lithuania	1
China	8	North America	2	Morocco	1
South Africa	7	Canada	2	Nigeria	1
Slovenia	6	Asia	2	Puerto Rico	1
Germany	5	Denmark	1	Scandinavia	1
France	3	Netherlands	1	Switzerland	1
Malaysia	3	Middle East	1	Austria	1
Europe	2	Ghana	1	Israel	1
Iran	2	Italy	1	Jordon	1
Peru	2	Korea	1	Brazil	1
Thailand	2	Bangladesh	1	Middle East	1

adoption was covered by 28 (28%) out of the total 111 studies (Fig. 5). These studies mainly focused on how BI systems are adopted by end-users while also identifying the factors that exert an impact on the adoption. For instance, Zhao, Navarrete and Iriberry [55] identified factors related to the organization (industry and size of resources), provider (vendor recognition, administration and deployment), project (team size and cost) and system (code, data and documentation quality) as being critical to the adoption of open-source BI tools. Mathew [56] investigated the factors associated with BI system adoption, namely

task characteristics (decision support and task complexity), retailer category (management and size), BI system provider characteristics (access and affordability) and decision-maker characteristics (technology, familiarity and quantitative skills).

The second category is BI system utilization, which has the second greatest number of studies, specifically 38% as shown in Fig. 5. Studies within this research category have provided a discussion on users' behaviours towards BI systems and the motivating factors that allow users to use or continue to use these systems. For example, Bischoff, Aier, Haki and Winter [57] adopted a mixed-method approach to explore the factors affecting BI systems continuous use. Their research revealed constructs such as governance, coverage of user requirements, influence of peers, influence of the organization, perceived ease of use and usefulness, user support and trust as the main drivers.

The third category is the BI systems success, highlighted in 41% of the studies. The studies within this research category focused on how organizations achieve success through the use of a BI system and mainly discussed success factors and outcomes, such as impacts, benefits and performance. Wieder, Ossimitz and Chamoni [40] identified factors such as BI management, data quality, BI scope, user satisfaction and BI use as being important for achieving benefits or success (decision quality and performance) from BI tools.

3.3.2. Key theories/frameworks/models adopted in BI system adoption, utilization and success research

The data analysis revealed that the selected studies employed a wide range of theories and models to examine BI system AUS. A total of 28 theories/frameworks/models were found covering the BI system AUS research domain, as shown in Table 11.

Among these theories/frameworks/models, DeLone and McLean's (D&M) IS success model, the technology acceptance model and the diffusion of innovation theory were found to be the most commonly used models in BI system AUS studies, as shown in Fig. 6.

DeLone and McLean's (D&M) IS success model [95,96] is one of the most-cited models to assess information technology success. The model proposes six IS success dimensions, namely information quality, service quality, system quality, use, user satisfaction and net benefits [95,96]. These dimensions cover the whole spectrum of information flows from the original production through consumption to the influence on individual and organizational performance. In the context of BI system research, the D&M model has remained the most influential framework in exploring BI system success. A total of 16 studies adopting this framework mainly discussed the factors critical to BI system success and the way in which BI systems influence individuals' decision-making performance and organizations' performance. For instance, a study conducted by Shin [68] applied the IS success framework to identify success measures in the business intelligence context. The study found that users' satisfaction is dependent on system quality factors, such as data locatability, data quality and system throughput. Mudzana and Maharaj [64] amplified the IS success framework as a means of investigating how quality factors, including system quality, information quality and service quality, contribute to the success of a BI system.

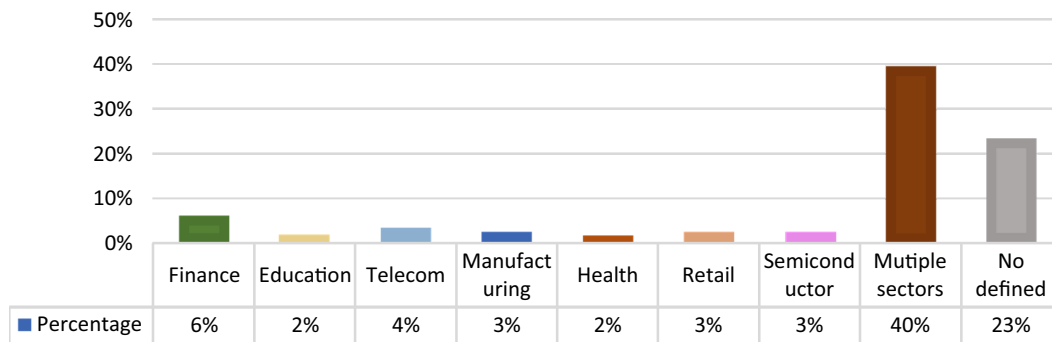


Fig. 4. Distribution of studies by sector.

Table 10
Research topic categorization.

Topic categorization	Description
Adoption	Definition: The initial BI system usage [51,52] Description: Studies explaining the adoption, initial BI system usage, users' intention and satisfaction and system evaluation
Utilization	Definition: The intensity of the BI system use [53,54] Description: Studies explaining the extent of BI system use and its validation
Success	Definition: A satisfactory/favourable outcome [53,54] Description: Studies explaining the benefits, effectiveness, impact and performance, value creation or outcomes

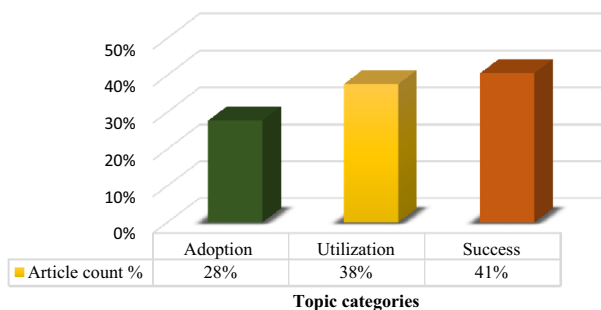


Fig. 5. BI system adoption, utilization & success – topic categories.

The second-most-adopted model in the BI system AUS research reviewed was the *technology acceptance model* (TAM), proposed by Davis [97]. TAM is considered one of the most famous and parsimonious models for evaluating individuals' IT acceptance. The model constitutes two key constructs, namely perceived usefulness (PU) and perceived ease of use (PEOU), which measure individuals' usage behaviour indirectly through behavioural intention. Fifteen studies have adopted the TAM to study perceptions of BI technology or the intention to accept or use BI technology [37,55,69]. For instance, Kohnke, Wolf and Mueller [37] applied the TAM to predict users' acceptance of a BI system. They found PEOU to be the strongest predictor of users' behavioural intention in the BI system context.

The next most-cited theory is *diffusion of innovation* (DOI), proposed by Rogers Everett [98]. According to DOI, innovation is communicated through different channels over time and within a particular social system [98]. The theory proposes five perceived characteristics of innovation adoption: compatibility, complexity, observability, relative advantage and trialability. The analysis revealed ten studies in total that used the DOI theory to explore BI technology adoption. For instance, Yoon, Ghosh and Jeong [7] and Boonsiritomachai, McGrath and Burgess [41] amplified DOI to investigate how relative advantage, compatibility, complexity and related factors affect BI technology adoption.

On the other hand, the adoption rate of other theories such as resource-based view (RBV), the unified theory of acceptance and use of technology (UTAUT), technology–organization–environment (TOE)

framework and the motivation theory (MT) was relatively low, as depicted in Fig. 6. Additionally, the absence of decision theory among the 111 BI AUS articles is surprising since a primary purpose of a BI system is to provide actionable intelligence to improve managerial decision-making. The application of decision theory to the effective use of information is crucial to the success of BI systems. This shortcoming is further discussed as an area for BI research improvement in section 5.

3.3.3. Key factors in BI system adoption, utilization and success

The selected studies were analysed further to identify the theoretical constructs or factors that influence BI system AUS. Taking into account the categorization method by Hwang, Ku, Yen and Cheng [99], the analysis revealed three main factor or driver categories, as shown in Table 12. The first category is the “organizational perspective” which focuses on how the alignment of organizational goals, strategies, plans and priorities with the BI systems affects system AUS. This category includes organization-related factors, such as management support [22], BI management [5] and technology-driven strategy [10]. The second category is the “information system (IS) perspective” which focuses on the importance of a scalable and flexible IT infrastructure, an easy-to-use system interface and a high-quality data and source system for BI system AUS. The “IS perspective” demonstrates the impact of IS-related factors, such as technological BI capabilities [114], information and system quality [64] and IT infrastructure [6], as listed in Table 12.

The third category is the “users' perspective” which shows the investigation of human-related factors, as shown in Table 12. The analysis of these key factors revealed that human factors have been largely ignored in the study of BI system AUS. Few studies have taken users' perspective into account. The focus of the majority of studies has remained limited to either organizational or IS-related factors as the most important influencers to AUS.

3.3.4. Challenges in BI system adoption, utilization and success

Undoubtedly, BI systems are critical for organizations because of their ability to predict, reason, plan and solve problems in a way that enhances organizational decision processes, enables effective actions and helps to attain organizational goals [65]. However, this study found that organizations face many challenges in terms of system adoption, usage and implementation success. A few challenges, highlighted in recent studies, are listed in Table 13. One important challenge to the

Table 11
Key theoretical lenses.

Category	Theory/model/frameworks	No. of times used	References
Success	DeLone & McLean IS success model	16	[5,23,29,36,40,58–68]
Adoption and utilization	Technology acceptance model	15	[10,36,37,55,61,69–78]
	Diffusion of innovation theory	10	[7,41,77–84]
	Resource based view	4	[20,80,85,86]
	Unified Theory of acceptance and use of technology (UTAUT)	3	[72,80,87]
	Motivational theory	3	[7,77,88]
	Technology, Organization, Environment (TOE)	3	[81,83,84]
	Information processing theory	2	[86,89]
	Expectation-confirmation Model of IS	2	[46,76]
	Bagozzi, Dholakia and Basuroy (BDB) model	2	[35,90]
	Theory of Planned Behavior (TPB)	2	[7,75]
	Gorry and Scott Morton framework of management information system	2	[21,47]
	Technology to Performance Chain (TPC) model	1	[91]
	Nomological net model	1	[62]
	Expectancy Theory	1	[35]
	Social exchange theory	1	[35]
	Theory of effective use	1	[92]
	Porter's value-chain activities framework	1	[93]
	Institutional theory	1	[50]
	Technology adoption model	1	[56]
	Theory of Reasoned action (TRA)	1	[77]
	Wixom & Watson framework	1	[22]
	Theory of effective use	1	[92]
	Limayem et al.'s IS continuance model	1	[94]
	Strategic orientation of business enterprise (STROBE) framework	1	[20]
	Burton-Jones and Straub dimensions	1	[6]
	Clark's model	1	[17]
	Integrative model of IT value	1	[89]

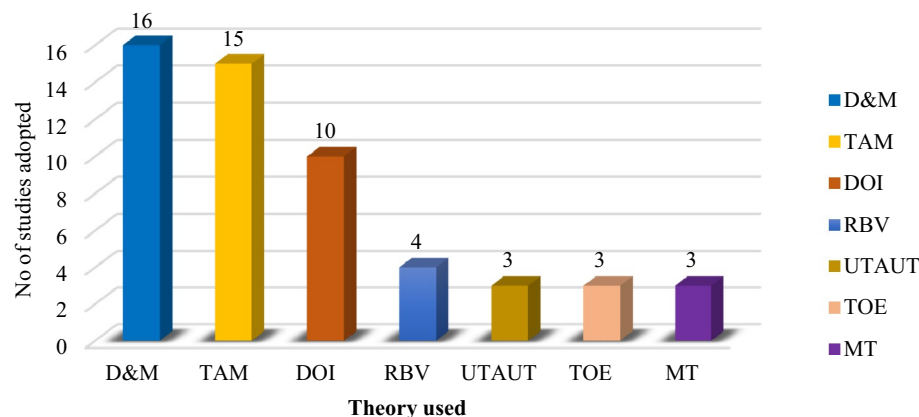


Fig. 6. Key theoretical lenses used in BI system research.

adoption of BI systems is “individual-level acceptance and use.” Users’ low level of acceptance or their resistance to utilizing BI systems is a key challenge for the management [35–38]. From this perspective, researchers have highlighted users’ lack of motivation, capabilities, ability to explore the system and system logics and system errors as key challenges at the user level [39,40].

Likewise, Popović [38] found that the fear of losing power over information, a change in job skills (e.g. the requirement of new skills to perform routine tasks) and a change in the decision-making approach (e.g. integrating BIS into organizational processes) are the main reasons for users’ resistance to a system.

BI systems support analytical decision making in knowledge-intensive activities. In this view, some challenges include a lack of knowledge about the system and the absence of (required) technical skills [41]. Users may be unwilling to embed a BI system into their routine tasks if they believe that they do not possess the knowledge and technical skills required to use that system.

Other researchers have pointed out that the use and success of a BI system are affected by infrastructural issues [42,43], insufficient

communication between IT staff and business users on the use of the system [44], the absence of an information culture, inappropriate training and insufficient service quality [38]. Deng and Chi [45] emphasized that problems associated with system use must be overcome by the management to integrate systems into their routine work and to exploit the full benefits of the implemented systems. A lack of timely responses to users’ difficulty in employing system features in their assigned tasks can limit the use of a BI system, which subsequently affects the task performance of both users and organizations negatively.

3.3.5. Knowledge gaps and prospects for future research

The examination of the selected studies revealed that the research on BI systems AUS has evolved gradually and prompted increased interest and attention among scholars and practitioners over the last decade. From this review’s findings, it is notable that the organizational and IS perspectives were considered frequently while investigating the adoption, utilization or success of BI systems while the user perspective was less frequently considered. Individual user acceptance and effective use is one of the greatest challenges for BI systems. An organization’s

Table 12
Categorization of key factors adopted in BI system AUS studies.

Perspectives	Categories	Key factors	Adoption	Utilization	Success
Organizational perspective—organizational related factors		Management support	[84,99]	[37,81,100]	[29,47,58,101]
		Champion	[83,84,99]		[22,29,48,58]
		Support and training	[28]	[38,57,61,100]	[101]
		Culture	[83,84]	[38]	[20,63,102]
		Social influence	[7,75,77,80]	[57,72,87]	
		Resources	[7,55]		[22,101]
		Change management	[10]	[100]	[48,101]
		Facilitating conditions	[28,75]	[72,73,87]	[58]
		Organization size	[55,78,83]		
		Service quality		[38]	[60,67,68]
		Well defined vision and goals, BI & business strategy alignment, effective communication, effective project management, teamwork & composition, agile values, plan driven aspects		[100]	[58,101,103]
		Competitive pressure	[41]	[50]	
		Structural empowerment		[46,94]	
		BI management			[5,40]
		Organizational data environment, organizational readiness, external support	[83,84]		
		User participation			[22,101]
IS perspective – IS related factors		Organizational learning climate	[7]		
		Organizational BI capabilities	[52]		
		Top management commitment			[103,104]
		Knowledge sharing, technology driven strategy	[10]		
		Information/data quality	[10,28,55,105,106]	[17,31,36,37,57,61,62,73,74,91,100,107]	[22,23,47,48,60,64,68,102,108–110]
		System quality	[55,71,105,106]	[6,91]	[23,60,64,68,109]
		Perceived ease of use	[70,75,77,105]	[69,74,76]	[108]
		Result demonstrability	[80]	[72–74]	
		Perceived usefulness	[75,77]	[46,76,88]	[108]
		Relative advantage	[41,84]		[79]
		Job relevance		[73,74]	
		BI system maturity, BIS effectiveness	[111]		[65,89]
		Comprehensiveness of usage		[46,94]	
		Compatibility	[7,75,80]		
		Performance expectancy, effort expectancy	[80]	[72,87]	[48,86,104,112]
		IT infrastructure, integration		[100]	[113]
Users perspective – human related factors		Information and analysis usage, Technical readiness of BI			[47]
		Integration with other systems, user access			[102]
		BIS dependence			[86]
		BIS infusion			
		Management capability, sensing capability, seizing capability, business process change capability			[86]
		Functional performance			[82]
		Impact on marketing & Sales, Impact on management and internal operations, Impact on procurement			
		Technological BI capabilities - data source, data type, data reliability, interaction with other systems, user access			[103,114]
		Anxiety		[73,87]	
		Absorptive capacity	[41]	[49]	
		Team IT knowledge and technical skills	[41]	[11]	[86,101]
		Self-efficacy	[75]	[87]	
		User involvement			[63]
		Personal innovativeness	[52,82]		
		Loss of power, changes in decision-making approach, job/skills change		[38]	
		Conscientiousness, emotional stability, extraversion, openness to experience		[90]	

Table 13
Challenges.

Challenges	Citations
- Low of system acceptance - Resistance to use BI systems	Chang, Hsu and Wu [35], [36–38]
- Lack of motivation - Lack of capabilities - Lack of ability to explore the system - Lack of system logics - System errors	Seah, Hsieh and Weng [39,40]
- Fear of losing power over information, - Change in job/job skills - Change in decision-making approach - Absence of information culture - Inappropriate training - Insufficient service quality	Popovič [38]
- Lack of knowledge - Absence of requisite technical skills	Boonsiritomachai, McGrath and Burgess [41]
- System issues - Infrastructural issues	Hannula and Pirttimäki [42], Olszak [43]
- Insufficient communication between IT staff and business users	Richards, Yeoh, Chong and Popovič [44]
- Lack of timely response, problems of reporting data, lack of knowledge, user-system interaction, system error	Deng and Chi [45]

goal of achieving high returns through BI investments is highly dependent on the effective utilization of a BI system [12], which in turn depends on the end-users. On the flip side, users' resistance or under-utilization of BI systems may result in workflow problems [12,38], that ultimately produce strategy blindness [115] and negative business performance [45]. Therefore, it is important to understand individual-level issues and resolve them to exploit the full benefits of BI systems and to reduce the risk of implementation failure [38,45].

However, knowledge gaps exist for three user level areas and therefore represent opportunities for future research; namely, individual IT competencies, user perceptions and user decision performance.

The first promising focus for future research would be user IT competencies, namely IT-related skills, IT knowledge, utilization ability or any other individual characteristics. Research on individual IT competencies is important, since BI systems comprise reporting and analytics for end-users. The former involves the creation of reports through dragging and dropping, whereas the latter involves business knowledge discovery and deep analysis with advanced statistical functions [35]. In the case of the presence of the required IT competencies (such as IT knowledge and skills), users would deploy a BI system in their routine tasks easily, while, in the case of their absence, users may have to invest more effort in gaining an understanding of the given functions and available data, which may lead them to avoid using the system. In addition, along with the IT competencies, the presence of specialized analytical skills such as skills to perform statistical, financial analysis and forecasting model building [116], would escalate the use

and success of BI system. Many researchers have highlighted the significance of users' competencies for the system use and success of IS [9,116,117] in general and BI systems specifically [37,44]. However, this review's findings revealed that none of the empirical studies have empirically tested IT competencies at the individual level in BI system AUS research contexts. Therefore, future BI system research should include users' IT competencies. Future studies could also apply individual-level theories, such as the motivation theory [118] and social cognitive theory [122] to determine individual behaviour in the BI system research domain.

Secondly, the exploration of users' perception is important, because users' positive perception of a BI system performance would improve their likelihood of using the system [6] and ultimately the success of the system. The perceived positive performance impact would encourage users to apply the system frequently to their routine tasks. This review's findings revealed that, with the exception of researchers applying the TAM model, studies have overlooked the perceived system impact in BI system AUS research. Thus, this study suggests that system perceived impact (e.g. task productivity, task innovation and management control) – proposed by Torkzadeh and Doll [119] – could be taken into consideration to gain a better understanding of users' perspective about the innovative and productive use of BI systems.

In addition, this review of the BI literature showed that the majority of BI research has focused on the adaptation of traditional IS theories to investigate BI system AUS. These studies have mainly focused individuals' system use and their performance, and organizational performance. According to Yuthas and Eining [120], when information is critical to improve managerial decision making, then decision making performance would be relevant for the evaluation of an information system. BI systems are solutions typically designed to support decision making in an organization [65], yet the measurement of individual's decision-making performance are largely missing from the BI system research. Thus, Huber's theory [121] could be applied in future research to understand the impact of advanced technologies such as BI technologies on organizational decision making in order to evaluate BI success. Table 14 presents a summary of past knowledge, gaps and future research prospects.

4. Limitation of the research

This SLR has several limitations that are important to note. These limitations in turn provide direction for future research. Firstly, the literature search was focused on three main categories of BI investigation, specifically the adoption, utilization and success of BI systems. Future research may broaden the research strategy to identify additional insights and opportunities in the BI system research domain. For example, future studies should focus on decision making performance as a key measure of BI success.

Secondly, the focus of the review was limited to BI systems and a limited set of their components, such as data warehouses, OLAP and dashboards. Future studies could consider data exploration techniques,

Table 14
Summary of existing knowledge, knowledge gaps and future research prospects.

	Existing body of knowledge	Knowledge gaps and prospect for future research
BI AUS research	<i>Organizational perspective</i> • Focused on organizational aspects such as management support [22], BI management [110], well-defined vision and goals [79] etc. <i>IS perspective</i> • Focused on IS aspects such as scalable and flexible IT infrastructure [37], data quality [11], compatibility [51] etc.	<i>Measurement of the users' perspective</i> • Users competencies – IT competencies and analytical skills. • Users perceptions towards BI system –perceived system impact at the individual level. • Users' decision-making performance
Theoretical groundings	<i>Adaptation of IS theories</i> • D&M (16) • TAM (15) • DOI (10) • Others [See Table 11]	<i>Individual-level theories</i> • Motivation theory • Social cognitive theory <i>Decision theory</i> • Huber's theory

for example, data mining, to provide additional useful insights.

Lastly, this review comprehensively examined the theoretical lenses that are the most influential in the field and the key factors or drivers that contribute to adoption, utilization and success. However, it did not consider the relationships among factors and their collective impact on success outcomes, such as the effectiveness and efficiency of the decision-making process. Future research could employ a meta-analysis to extend knowledge in this area.

5. Implications and conclusions

This SLR provides contributions for both academic scholars and practitioners. For scholars, it offers a valuable synthesis of 20 years of BI research to better understand the current state of BI system AUS research. BI system AUS research is still progressing, and this review can serve as a reference for scholars, as it reports on the theoretical lenses, key factors and research methodologies used in the BI system AUS studies to date. Additionally, the review provides a rich picture of the existing literature, informing scholars of the areas in which the research is lacking and further exploration is needed. The results of the review will also be useful for scholars to direct their future efforts in research that might be more applicable to organizations in terms of helping them to improve the use and success of the implemented BI systems.

Researchers can apply one of the three most frequently used research frameworks or theories (DeLone & McLean IS Success Model, TAM or DOI) as the foundation for their BI system research. However, the study of BI system success needs to be grounded also on decision theories to help understand how systems and the information they produce can benefit the decision process. In addition, scholars can apply and integrate individual-level theories to reflect a more comprehensive view of BI system AUS. Researchers should devote more attention to exploring the impact of user-centred factors, such as IT competencies on BI system AUS.

For practitioners, such as business analysts, managers and IT executives, the analysis of the BI system AUS literature provides some needed insights. Despite increasing investments in BI systems, many organizations are still unable to attain the desired success from these systems due to underutilization and ineffective use [15,20]. Our analysis of the past two decades of research on BI system AUS has revealed that management support, training, the organizational culture, well-defined vision and goals, BI & business strategy alignment, data and system quality, and IT infrastructure as being important to BI AUS. Organizations also need to pay more attention to user-centred issues to improve the success of their BI system investments. BI AUS is dependent on users [43], so organizations must emphasize the development of specific capabilities and competencies (of users) to realize organizational success. Additionally, a corporate culture based on facts, knowledge and learning can motivate organizations to apply the information offered by BI systems.

In conclusion, this study has provided a comprehensive and systematic review of BI system AUS research over the period of the last two decades. Following a rigorous guideline for studies' selection, 111 peer-reviewed studies were identified and reviewed. The results obtained from the review indicated that BI research covering three main categories, adoption, utilization and success, made significant progress from 2000 to 2019. The distribution of the selected studies with regard to the different sectors/industries showed that most of the BI system research was conducted in the government services, transportation, insurance, communications, health care, banking, agriculture, construction and professional services sectors. Additionally, the data analysis with regard to the research methodologies revealed that the quantitative methods remained the most-adopted research approach. Furthermore, frameworks such as the DeLone & McLean IS success model, technology acceptance model and diffusion of innovation theory were the most dominant in BI system research.

Researchers have identified different factors that are critical to BI

system AUS. A close examination of the key factors revealed that the majority of studies reported either organizational or IS-related factors as being the most important to AUS, whereas user-related measures have received limited attention. Based on the review of 111 studies, the study also highlighted 1) the challenges facing BI systems, 2) gaps in our knowledge of BI systems AUS and 3) prospects for future research.

References

- [1] H. Chen, R.H. Chiang, V.C. Storey, Business intelligence and analytics: from big data to big impact, *MIS Quarterly* (2012) 1165–1188.
- [2] B. Wixom, H. Watson, The BI-based organization, *International Journal of Business Intelligence Research (IJIBIR)* 1 (1) (2010) 13–28.
- [3] S. Chaudhuri, U. Dayal, V. Narasayya, An overview of business intelligence technology, *Communications of the ACM* 54 (8) (2011) 88–98.
- [4] R. Harrison, A. Parker, G. Brosas, R. Chiong, X. Tian, The role of technology in the management and exploitation of internal business intelligence, *Journal of Systems and Information Technology* 17 (3) (2015) 247–262.
- [5] B. Wieder, M.-L. Ossimitz, The impact of business intelligence on the quality of decision making—a mediation model, *Procedia Computer Science* 64 (2015) 1163–1171.
- [6] T. Grublješić, J. Jaklić, Conceptualization of the business intelligence extended use model, *Journal of Computer Information Systems* 55 (3) (2015) 72–82.
- [7] T.E. Yoon, B. Ghosh, B.-K. Jeong, User acceptance of business intelligence (BI) application: technology, individual difference, social influence, and situational constraints, *System Sciences (HICSS)*, 2014 47th Hawaii International Conference on, IEEE, 2014, pp. 3758–3766.
- [8] T.D. Clark, M.C. Jones, C.P. Armstrong, The dynamic structure of management support systems: theory development, research focus, and direction, *MIS Quarterly* 31 (3) (2007) 579–615.
- [9] C.Y. Yoon, A structural model of end-user computing competency and user performance, *Knowledge-Based Systems* 21 (5) (2008) 415–420.
- [10] M.P. Bach, A. Celjo, J. Zoroja, Technology acceptance model for business intelligence systems: preliminary research, *Procedia Computer Science* 100 (2016) 995–1001.
- [11] M.Z. Elbashir, P.A. Collier, S.G. Sutton, M.J. Davern, S.A. Leech, Enhancing the business value of business intelligence: the role of shared knowledge and assimilation, *Journal of Information Systems* 27 (2) (2013) 87–105.
- [12] V.-H. Trieu, Getting value from business intelligence systems: a review and research agenda, *Decision Support Systems* 93 (2017) 111–124.
- [13] Gartner, Business Intelligence and Analytics Trends in, (2017).
- [14] A. Audzeyeva, R. Hudson, How to get the most from a business intelligence application during the post implementation phase? Deep structure transformation at a UK retail bank, *European Journal of Information Systems* 25 (1) (2016) 29–46.
- [15] Gartner, Business Intelligence & Analytics Summit in, (2015).
- [16] W. Yeoh, A. Popović, Extending the understanding of critical success factors for implementing business intelligence systems, *Journal of the Association for Information Science and Technology* 67 (1) (2016) 134–147.
- [17] L.L. Visinescu, M.C. Jones, A. Sidorova, Improving decision quality: the role of business intelligence, *Journal of Computer Information Systems* 57 (1) (2017) 58–66.
- [18] Z. Jourdan, R.K. Rainer, T.E. Marshall, Business intelligence: an analysis of the literature, *Information Systems Management* 25 (2) (2008) 121–131.
- [19] R. Fitriana, T.D. Eriyatno, T. Djatna, Progress in business intelligence system research: a literature review, *International Journal of Basic & Applied Sciences IJBAS-IJENS* 11 (3) (2011) (118503-116464).
- [20] M.S. Arefin, M.R. Hoque, Y. Bao, The impact of business intelligence on organization's effectiveness: an empirical study, *Journal of Systems and Information Technology* 17 (3) (2015) 263.
- [21] D. Arnott, F. Lizama, Y. Song, Patterns of business intelligence systems use in organizations, *Decision Support Systems* 97 (2017) 58–68.
- [22] L. Dawson, J.-P. Van Belle, Critical success factors for business intelligence in the South African financial services sector, *South African Journal of Information Management* 15 (1) (2013) 1–12.
- [23] R. Gaardboe, T. Nyvang, N. Sandalgaard, Business intelligence success applied to healthcare information systems, *Procedia Computer Science* 121 (2017) 483–490.
- [24] B. Kitchenham, Procedures for Performing Systematic Reviews, 33(2004) Keele University, Keele, UK, 2004, pp. 1–26.
- [25] P. Brereton, B.A. Kitchenham, D. Budgen, M. Turner, M. Khalil, Lessons from applying the systematic literature review process within the software engineering domain, *Journal of Systems and Software* 80 (4) (2007) 571–583.
- [26] B. Kitchenham, S. Charters, Guidelines for Performing Systematic Literature Reviews in Software Engineering, V 2 (2007), pp. 1–65.
- [27] J. Webster, R.T. Watson, Analyzing the past to prepare for the future: writing a literature review, *MIS Quarterly* (2002) xiii–xxiii.
- [28] K.S. Soliman, E. Mao, M.N. Frolick, Measuring user satisfaction with data warehouses: an exploratory study, *Information & Management* 37 (3) (2000) 103–110.
- [29] B.H. Wixom, H.J. Watson, An empirical investigation of the factors affecting data warehousing success, *MIS Quarterly* (2001) 17–41.
- [30] S. Nidhra, M. Yanamadala, W. Afzal, R. Torkar, Knowledge transfer challenges and mitigation strategies in global software development—a systematic literature review and industrial validation, *International Journal of Information Management* 33 (2) (2013) 333–355.

- [31] D.A. Adams, R.R. Nelson, P.A. Todd, Perceived usefulness, ease of use, and usage of information technology: a replication, *MIS Quarterly* (1992) 227–247.
- [32] H. Hoehle, E. Scornavacca, S. Huff, Three decades of research on consumer adoption and utilization of electronic banking channels: a literature analysis, *Decision Support Systems* 54 (1) (2012) 122–132.
- [33] B. Kitchenham, O.P. Brereton, D. Budgen, M. Turner, J. Bailey, S. Linkman, Systematic literature reviews in software engineering—a systematic literature review, *Information and Software Technology* 51 (1) (2009) 7–15.
- [34] B. Kitchenham, R. Pretorius, D. Budgen, O.P. Brereton, M. Turner, M. Niazi, S. Linkman, Systematic literature reviews in software engineering—a tertiary study, *Information and Software Technology* 52 (8) (2010) 792–805.
- [35] Y.-W. Chang, P.-Y. Hsu, Z.-Y. Wu, Exploring managers' intention to use business intelligence: the role of motivations, *Behaviour & Information Technology* 34 (3) (2015) 273–285.
- [36] N. Foshay, A. Taylor, A. Mukherjee, Winning the hearts and minds of business intelligence users: the role of metadata, *Information Systems Management* 31 (2) (2014) 167–180.
- [37] O. Kohnke, T.R. Wolf, K. Mueller, Managing user acceptance: an empirical investigation in the context of business intelligence standard software, *International Journal of Information Systems and Change Management* 5 (4) (2011) 269–290.
- [38] A. Popović, If we implement it, will they come? User resistance in post-acceptance usage behaviour within a business intelligence systems context, *Economic Research-Ekonomska Istraživanja* 30 (1) (2017) 911–921.
- [39] M. Seah, M.H. Hsieh, P.-D. Weng, A case analysis of Savecom: the role of indigenous leadership in implementing a business intelligence system, *International Journal of Information Management* 30 (4) (2010) 368–373.
- [40] B. Wieder, M. Ossimitz, P. Chamoni, The Impact of Business Intelligence Tools on Performance: A User Satisfaction Paradox? (2012).
- [41] W. Boonsiritomachai, G.M. McGrath, S. Burgess, Exploring business intelligence and its depth of maturity in Thai SMEs, *Cogent Business & Management* 3 (1) (2016) 1220663.
- [42] M. Hannula, V. Pirttimäki, Business intelligence empirical study on the top 50 Finnish companies, *Journal of American Academy of Business* 2 (2) (2003) 593–599.
- [43] C.M. Olszak, Toward better understanding and use of business intelligence in organizations, *Information Systems Management* 33 (2) (2016) 105–123.
- [44] G. Richards, W. Yeoh, A.Y.L. Chong, A. Popović, Business intelligence effectiveness and corporate performance management: an empirical analysis, *Journal of Computer Information Systems* (2017) 1–9.
- [45] X. Deng, L. Chi, Understanding postadoptive behaviors in information systems use: a longitudinal analysis of system use problems in the business intelligence context, *Journal of Management Information Systems* 29 (3) (2012) 291–326.
- [46] Y.-M. Han, C.-S. Shen, C.-K. Farn, Determinants of continued usage of pervasive business intelligence systems, *Information Development* 32 (3) (2016) 424–439.
- [47] Ö. Işık, M.C. Jones, A. Sidorova, Business intelligence success: the roles of BI capabilities and decision environments, *Information & Management* 50 (1) (2013) 13–23.
- [48] W. Yeoh, A. Koronios, Critical success factors for business intelligence systems, *Journal of Computer Information Systems* 50 (3) (2010) 23–32.
- [49] M.Z. Elbashir, P.A. Collier, S.G. Sutton, The role of organizational absorptive capacity in strategic use of business intelligence to support integrated management control systems, *The Accounting Review* 86 (1) (2011) 155–184.
- [50] T. Ramakrishnan, M.C. Jones, A. Sidorova, Factors influencing business intelligence (BI) data collection strategies: an empirical investigation, *Decision Support Systems* 52 (2) (2012) 486–496.
- [51] E. Karahanna, D.W. Straub, N.L. Chervany, Information technology adoption across time: a cross-sectional comparison of pre-adoption and post-adoption beliefs, *MIS Quarterly* (1999) 183–213.
- [52] H.-C. Wang, Distinguishing the adoption of business intelligence systems from their implementation: the role of managers' personality profiles, *Behaviour & Information Technology* 33 (10) (2014) 1082–1092.
- [53] C.-K. Hou, K.N. Papamichail, The impact of integrating enterprise resource planning systems with business intelligence systems on decision-making performance: an empirical study of the semiconductor industry, *International Journal of Technology, Policy and Management* 10 (3) (2010) 201–226.
- [54] E.W. Ngai, J. Poon, Y.H. Chan, Empirical examination of the adoption of WebCT using TAM, *Computers & Education* 48 (2) (2007) 250–267.
- [55] Z. Zhao, C. Navarrete, A. Iriberrí, Open Source Alternatives for Business Intelligence: Critical Success Factors for Adoption, (2012).
- [56] S.K. Mathew, Adoption of business intelligence systems in Indian fashion retail, *International Journal of Business Information Systems* 9 (3) (2012) 261–277.
- [57] S. Bischoff, S. Aier, M.K. Haki, R. Winter, Understanding continuous use of business intelligence systems: a mixed methods investigation, *JITTA: Journal of Information Technology Theory and Application* 16 (2) (2015) 5.
- [58] I. Candal-Vicente, Factors that affect the successful implementation of a data warehouse, *Computing in the Global Information Technology*, 2009. ICCGI'09. Fourth International Multi-Conference on, IEEE, 2009, pp. 1–6.
- [59] N. Dedić, C. Stanier, Measuring the success of changes to business intelligence solutions to improve business intelligence reporting, *Journal of Management Analytics* 4 (2) (2017) 130–144.
- [60] R. Gonzales, J. Wareham, J. Serida, Measuring the impact of data warehouse and business intelligence on enterprise performance in Peru: a developing country, *Journal of Global Information Technology Management* 18 (3) (2015) 162–187.
- [61] S. Hong, P. Katerattanakul, S.-K. Hong, Q. Cao, Usage and perceived impact of data warehouses: a study in Korean financial companies, *International Journal of Information Technology & Decision Making* 5 (2) (2006) 297–315.
- [62] C.-K. Hou, Examining the effect of user satisfaction on system usage and individual performance with business intelligence systems: an empirical study of Taiwan's electronics industry, *International Journal of Information Management* 32 (6) (2012) 560–573.
- [63] U. Kulkarni, J.A. Robles-Flores, Development and Validation of a BI Success Model, (2013).
- [64] T. Mudzana, M. Maharaj, Measuring the success of business-intelligence systems in South Africa: an empirical investigation applying the DeLone and McLean model, *South African Journal of Information Management* 17 (1) (2015) 1–7.
- [65] A. Popović, R. Hackney, P.S. Coelho, J. Jaklič, Towards business intelligence systems success: effects of maturity and culture on analytical decision making, *Decision Support Systems* 54 (1) (2012) 729–739.
- [66] C. Schieder, P. Gluchowski, Towards a Consolidated Research Model for Understanding Business Intelligence Success, *ECIS*, 2011, p. 205.
- [67] P.A. Serumaga-Zake, The role of user satisfaction in implementing a business intelligence system, *South African Journal of Information Management* 19 (1) (2017).
- [68] B. Shin, An exploratory investigation of system success factors in data warehousing, *Journal of the Association for Information Systems* 4 (1) (2003) 6.
- [69] T. Brockmann, S. Stieglitz, J. Kmiecik, S. Diederich, User acceptance of mobile business intelligence services, *Network-Based Information Systems (NBIS)*, 2012 15th International Conference on, IEEE, 2012, pp. 861–866.
- [70] Y.-W. Chang, P.-Y. Hsu, W.-L. Shiau, An empirical study of managers' usage intention in BI, *Cognition, Technology & Work* 16 (2) (2014) 247–258.
- [71] N. Gorla, Features to consider in a data warehousing system, *Communications of the ACM* 46 (11) (2003) 111–115.
- [72] T. Grublješić, P.S. Coelho, J. Jaklič, The importance and impact of determinants influencing business intelligence systems embeddedness, *Issues in Information Systems* 15 (1) (2014).
- [73] M. Hart, F. Esat, M. Rocha, Z. Khatieb, Introducing students to business intelligence: acceptance and perceptions of OLAP software, *Issues in Informing Science & Information Technology* 4 (2007).
- [74] M. Hart, G. Porter, The impact of cognitive and other factors on the perceived usefulness of OLAP, *Journal of Computer Information Systems* 45 (1) (2004) 47–56.
- [75] C.-K. Hou, Investigating factors influencing the adoption of business intelligence systems: an empirical examination of two competing models, *International Journal of Technology, Policy and Management* 13 (4) (2013) 328–353.
- [76] C.K. Hou, Understanding business intelligence system continuance intention an empirical study of Taiwan's electronics industry, *Information Development* 32 (5) (2016) 1359–1371.
- [77] Y. Jiang, A conceptual framework and hypotheses for the adoption of e-business intelligence, *Computing, Communication, Control, and Management*, 2009. CCCM 2009. ISECS International Colloquium on, IEEE, 2009, pp. 558–561.
- [78] K.R. Ramamurthy, A. Sen, A.P. Sinha, An empirical investigation of the key determinants of data warehouse adoption, *Decision Support Systems* 44 (4) (2008) 817–841.
- [79] A. Ahmad, R. Ahmad, K.F. Hashim, Innovation traits for business intelligence successful deployment, *Journal of Theoretical & Applied Information Technology* 89 (1) (2016).
- [80] J. Jaklič, T. Grublješić, A. Popović, The role of compatibility in predicting business intelligence and analytics use intentions, *International Journal of Information Management* 43 (2018) 305–318.
- [81] P. Lautenbach, K. Johnston, T. Adeniran-Ogundipe, Factors influencing business intelligence and analytics usage extent in South African organisations, *South African Journal of Business Management* 48 (3) (2017) 23–33.
- [82] A. Popović, B. Puklavec, T. Oliveira, Justifying business intelligence systems adoption in SMEs: impact of systems use on firm performance, *Industrial Management & Data Systems* 119 (1) (2019) 210–228.
- [83] B. Puklavec, T. Oliveira, A. Popovic, Unpacking business intelligence systems adoption determinants: an exploratory study of small and medium enterprises, *Economic and Business Review for Central and South-Eastern Europe* 16 (2) (2014) 185.
- [84] B. Puklavec, T. Oliveira, A. Popović, Understanding the determinants of business intelligence system adoption stages: an empirical study of SMEs, *Industrial Management & Data Systems*, (Just-Accepted) 118 (1) (2017) 236–261.
- [85] L. Fink, N. Yogev, A. Even, Business intelligence and organizational learning: an empirical investigation of value creation processes, *Information & Management* 54 (1) (2017) 38–56.
- [86] R. Torres, A. Sidorova, M.C. Jones, Enabling firm performance through business intelligence and analytics: a dynamic capabilities perspective, *Information & Management* 55 (7) (2018) 822–839.
- [87] C.-K. Hou, Exploring the user acceptance of business intelligence systems in Taiwan's electronics industry: applying the UTAUT model, *International Journal of Internet and Enterprise Management* 8 (3) (2014) 195–226.
- [88] X. Li, J.P.-A. Hsieh, A. Rai, Motivational differences across post-acceptance information system usage behaviors: an investigation in the business intelligence systems context, *Information Systems Research* 24 (3) (2013) 659–682.
- [89] G. Richards, W. Yeoh, A.Y.L. Chong, A. Popović, Business intelligence effectiveness and corporate performance management: an empirical analysis, *Journal of Computer Information Systems* 59 (2) (2019) 188–196.
- [90] Y.-W. Chang, P.-Y. Hsu, W.-L. Shiau, Z.-Y. Wu, The effects of personality traits on business intelligence usage: a decision-making perspective, *Malaysian Journal of Library & Information Science* 20 (2) (2017).
- [91] B. Kositanurit, K.-M. Osei-Bryson, O. Ngenyama, Re-examining information systems user performance: using data mining to identify properties of IS that lead

- to highest levels of user performance, *Expert Systems with Applications* 38 (6) (2011) 7041–7050.
- [92] T. Trieu, Extending the Theory of Effective Use: The Impact of Enterprise Architecture Maturity Stages on the Effective Use of Business Intelligence Systems, (2013).
- [93] M.Z. Elbashir, P.A. Collier, M.J. Davern, Measuring the effects of business intelligence systems: the relationship between business process and organizational performance, *International Journal of Accounting Information Systems* 9 (3) (2008) 135–153.
- [94] Y. Han, C. Farn, A study on the effects of empowerment and habit on continuance usage of pervasive business intelligence systems, *System Sciences (HICSS)*, 2013 46th Hawaii International Conference on, IEEE, 2013, pp. 3768–3777.
- [95] W.H. DeLone, E.R. McLean, Information systems success: the quest for the dependent variable, *Information Systems Research* 3 (1) (1992) 60–95.
- [96] W.H. DeLone, E.R. McLean, The DeLone and McLean model of information systems success: a ten-year update, *Journal of Management Information Systems* 19 (4) (2003) 9–30.
- [97] F.D. Davis, Perceived usefulness, perceived ease of use, and user acceptance of information technology, *MIS Quarterly* (1989) 319–340.
- [98] M. Rogers Everett, *Diffusion of Innovations*, New York, (1995), p. 12.
- [99] H.-G. Hwang, C.-Y. Ku, D.C. Yen, C.-C. Cheng, Critical factors influencing the adoption of data warehouse technology: a study of the banking industry in Taiwan, *Decision Support Systems* 37 (1) (2004) 1–21.
- [100] M.I.M. Nofal, Z.M. Yusof, Conceptual model of enterprise resource planning and business intelligence systems usage, *International Journal of Business Information Systems* 21 (2) (2016) 178–194.
- [101] A.Z. Ravasan, S.R. Savoji, An investigation of BI implementation critical success factors in Iranian context, *International Journal of Business Intelligence Research (IJBIR)* 5 (3) (2014) 41–57.
- [102] V.-H.T. Trieu, S. Cockcroft, A. Perdana, Decision-making performance in big data era: the role of actual business intelligence systems use and affecting external constraints, *Decision-Making* 11 (2018) 29–2018.
- [103] D. Batra, Agile values or plan-driven aspects: which factor contributes more toward the success of data warehousing, business intelligence, and analytics project development? *Journal of Systems and Software* 146 (2018) 249–262.
- [104] M.L. Gonzales, S. Mukhopadhyay, K. Bagchi, L. Gemoets, Factors influencing business intelligence-enabled success in global companies: an empirical study, *International Journal of Business Information Systems* 30 (3) (2019) 324–347.
- [105] S. Bouchana, M.A.J. Idrissi, Towards an assessment model of end user satisfaction and data quality in business intelligence systems, 2015 10th International Conference on Intelligent Systems: Theories and Applications (SITA), IEEE, 2015, pp. 1–6.
- [106] R.R. Nelson, P.A. Todd, B.H. Wixom, Antecedents of information and system quality: an empirical examination within the context of data warehousing, *Journal of Management Information Systems* 21 (4) (2005) 199–235.
- [107] B.H. Wixom, H.J. Watson, T. Werner, Developing an enterprise business intelligence capability: the Norfolk Southern journey, *MIS Quarterly Executive* 10 (2) (2011).
- [108] R. Masa'Deh, Z. Obeidat, M. Maqableh, M. Shah, The impact of business intelligence systems on an organization's effectiveness: the role of metadata quality from a developing country's view, *International Journal of Hospitality & Tourism Administration* (2018) 1–21.
- [109] A. Popović, R. Hackney, P.S. Coelho, J. Jaklič, How information-sharing values influence the use of information systems: an investigation in the business intelligence systems context, *The Journal of Strategic Information Systems* 23 (4) (2014) 270–283.
- [110] N. Zellal, A. Zaouia, An exploratory investigation of factors influencing data quality in data warehouse, *Complex Systems (WCCS)*, 2015 Third World Conference on, IEEE, 2015, pp. 1–6.
- [111] R. Skyrius, I. Katin, M. Kazimianec, S. Nemitko, G. Rumšas, R. Žilinskas, Factors driving business intelligence culture, *Issues in Informing Science and Information Technology* 13 (unknown) (2016) 171–186.
- [112] M.D. Peters, B. Wieder, S.G. Sutton, J. Wakefield, Business intelligence systems use in performance measurement capabilities: implications for enhanced competitive advantage, *International Journal of Accounting Information Systems* 21 (2016) 1–17.
- [113] A.J. Karim, The value of competitive business intelligence system (CBIS) to stimulate competitiveness in global market, *International Journal of Business and Social Science* 2 (19) (2011).
- [114] S. Kokin, T.N. Wang, Business intelligence: relationship between capabilities, success and decision environment, *Applied Mechanics and Materials*, Trans Tech Publ, 2014, pp. 4361–4365.
- [115] V. Arvidsson, J. Holmström, K. Lyytinen, Information systems use as strategy practice: a multi-dimensional view of strategic information system implementation and use, *The Journal of Strategic Information Systems* 23 (1) (2014) 45–61.
- [116] G. Torkzadeh, J. Lee, Measures of perceived end-user computing skills, *Information & Management* 40 (7) (2003) 607–615.
- [117] H.-P. Shih, Assessing the effects of self-efficacy and competence on individual satisfaction with computer use: an IT student perspective, *Computers in Human Behavior* 22 (6) (2006) 1012–1026.
- [118] E.L. Deci, R.M. Ryan, *Handbook of Self-determination Research*, University Rochester Press, 2002.
- [119] G. Torkzadeh, W.J. Doll, The development of a tool for measuring the perceived impact of information technology on work, *Omega* 27 (3) (1999) 327–339.
- [120] K. Yuthas, M.M. Eining, An experimental evaluation of measurements of information system effectiveness, *Journal of Information Systems* 9 (2) (1995) 69–84.
- [121] G.P. Huber, A theory of the effects of advanced information technology, *academy of management, The Academy of Management Review* 15 (1) (1990) 47.
- [122] B. Albert, *Social Foundations of Thought and Action: A Social Cognitive Theory*, Prentice-Hall, Inc, NJ, US, 1986.

NoorUl Ain is a Ph.D scholar at the department of Management, Ca' Foscari University of Venice, Italy. Her research interest includes information system adoption and use, information sciences, social capital, digital learning, human resource management, competitive intelligence and business intelligence system. Her research work has been published in *Information Development*, *Behaviour & Information Technology* and *International Journal of Emerging Science*. Previously, she worked as a research associate at University of Malaya and also performed administrative duties in the IT department of Pakistan Telecommunication Limited.

Giovanni Vaia is a Professor of Digital management at the department of Management, Ca' Foscari University of Venice, Italy. He is also a director of the Digital Enterprise Lab at Ca' Foscari University of Venice. His research interest involves economics and management of digital innovation, governance, sourcing and organizational design. His research work has been published in *Journal of information technology* and *MIS Quarterly Executive*.

William H. DeLone is a Professor Emeritus of Information Technology (IT) at the Kodak School of Business at American University, Washington, DC. His primary areas of research include the assessment of information systems' effectiveness, digital governance, cybersecurity governance, and e-government & public value. He has been published in various information systems journals including *Information Systems Research*, *Management Information Systems Quarterly*, the *Journal of Management Information Systems*, the *Journal of the Association of Information Systems*, the *European Journal of Information Systems*, *IEEE Transactions on Software Engineering*, the *Journal of Information Technology*, the *Communications of the ACM*, *MIS Quarterly Executive*, *DataBase*, the *International Journal of Electronic Commerce*, and *Information Technology & People*. He received a BS degree in mathematics from Villanova University, an MS degree in industrial administration from Carnegie-Mellon University, and a PhD degree in computer information systems from the University of California Los Angeles.

Mehwish Waheed is an Alexander Von Humboldt research fellow. She is working in Technische Universität, Dortmund, Germany. Her research interests focus on knowledge quality in digital Learning, innovation adoption, innovative behaviour and information system. Her research work publishes in *Journal of Computer Assisted Learning*, *Internet Research*, *Behaviour & Information Technology*, *Serial Reviews* and *Information Development*. She is actively involved as a reviewer for *Interactive Learning Environment*, *Computers in Human Behaviour*, and *Internet Research*. Prior, she served as eLearning Administrator at Allama Iqbal Open University, as a lecturer at International Islamic University and as a research associate at Asia Europe Institute, University of Malaya, Malaysia.